

Data Structure and Programming Spring 2025

Programming Assignment #4

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Release Date: 12:00 on 05/08/2025

Due Date: 23:59 on 05/22/2025

Late policy: $\text{Score} = \text{Score} \times 0.7^{(h/24)}$, $h = \text{late hours} \leq 48$

1 Problem Description

The median is the middle value in an ordered integer list. If the size of the list is even, there is no middle value, and the median is the mean of the two middle values.

- For example, for `array = [2,3,4]`, the median is 3.
- For example, for `array = [2,3]`, the median is $(2 + 3)/2 = 2.5$.

In this homework, you are expected to implement both `MinHeap` and `MaxHeap` classes, and utilize them to further implement a `FindMedian` class.

2 Evaluation

In this homework, you are expected to implement two fundamental data structures: `MinHeap` and `MaxHeap`. You will also use them to implement a `FindMedian` class that efficiently computes the median from a stream of numbers.

1. Please only modify the `TODO` sections in the provided code template. Do not change any other parts of the code.
2. Do not modify the interface of the functions, but you can add your own functions.
3. Any statistics library is not allowed.
4. You should not use the `heapq` library.
5. The time complexity of `AddNewValues` method for a single item should be $O(\log n)$, `RemoveMedian` method should be $O(\log n)$, and `ShowMedian` method should be $O(1)$.

3 Testing Instructions

To test your program, we provide a test file `main.py`. Please use the following command:

```
python3 main.py --input input_0.txt
--output0 output_min_0.txt
--output1 output_max_0.txt
--output2 output_median_0.txt
--golden0 golden_min_0.txt
--golden1 golden_max_0.txt
--golden2 golden_median_0.txt
--print 1 [--mode 0|1|2]
```

You may change the filenames (e.g., `input_0.txt`, `output_min_0.txt`, etc.) according to your test cases. The parameters are described as follows:

- `--input`: The input test file, containing the list of numbers.
- `--output0`: Output file for the **min heap**.
- `--output1`: Output file for the **max heap**.
- `--output2`: Output file for the **median values** derived from the stream.
- `--golden0`, `--golden1`, `--golden2`: Golden output files, which contain the expected results for comparison.
- `--print`: Controls the amount of printed debug information for **MinHeap** and **MaxHeap**.
 - 1: Print the heap status after **every insertion**, and once after the **removal**. Use this mode to verify correctness.
 - 0: Only print the heap **once after all insertions**, and once after the **removal**. Use this mode to measure execution time.
- `--mode` (optional): Specify which module to test.
 - 0: Test **MinHeap** only.
 - 1: Test **MaxHeap** only.
 - 2: Test **FindMedian** only.
 - (If omitted, all three will be tested.)

Note: If you run the script without any arguments: `python3 main.py`

It will be equivalent to running the full command with the upper default settings. If your environment doesn't support `python3`, you can use `python` instead.

4 Grading Policy

This programming assignment will be graded based on correctness and time complexity. We provide you one public test case: `input0.txt`, which contains 5 test inputs.

- **Public Test Case (`input0.txt`)**: 40% of the total score. This test case will primarily assess the correctness of your program with small input sizes.
- **Private Test Cases**: 60% of the total score. These private test cases will assess both the correctness and time efficiency of your program. They consist of small and large input sizes to test the program's scalability and performance.

The total points will be 100.

5 Submission

Please put `Median.py` into a directory named `studentID` and compress the directory into `studentID.zip` to NTUCOOL. The homework is due on 05/22, at 23:59.

- Please ensure that your code compiles correctly. You CANNOT ask the TA to debug or modify your code for evaluation.
- We will unzip your submission by running `unzip *.zip`. Your submission folder structure must be valid, or you will receive a 20% penalty for an incorrect format. Please follow the structure shown below carefully:

```
b10901143.zip/  
  b10901143/  
    Median.py
```